Low (2005): Self-Insurance in a Life-Cycle Model of Labour Supply and Savings

April 2, 2026

Hamish W. Low[1]

Abstract

This paper analyses the incentives to work and to save over the life cycle in the presence of incomplete markets. In a calibrated, partial equilibrium model, flexibility in hours worked changes asset age-profiles: borrowing when young is greater and saving when middle-aged is greater than when labour supply is fixed. Uncertainty causes individuals to work longer hours and to consume less when young. With flexibility over hours, accumulating precautionary assets incurs less of a utility cost and so the level of saving is greater. Further, allowing for flexibility and uncertainty means simulated hours of work and consumption more closely match the age profiles in the data.

Keywords Precautionary saving; Life-cycle labour supply

JEL codes D91; H31

Powered by Econ-ARK

```
GitHub Repo:        https://github.com/jzhan476/Low2005
             html:  https://jzhan476.github.io/Low2005 /
              PDF:  Low2005.pdf

Replication Package:10.5281/zenodo.17861977

Other Materials:
 Replication code:  Code/Python
```

Forthcoming, *Quantitative Economics*

The authors would like to thank seminar participants at the Board of Governors of the Federal Reserve System, Deutsche Bundesbank, the European Central Bank, the Norwegian University of Science and Technology, Oslo Macro Group, the Society for Computational Economics in 2023 and 2024, Statistics Norway, the University of Pennsylvania, Vanderbilt University, Johns Hopkins University, and West Virginia University for valuable feedback. The authors would also like to thank Sasha Indarte, Adrian Monninger, Plutarchos Sakellaris, three anonymous referees, and the editor Morten Ravn for insightful comments on earlier versions of the paper. Thanks also to the hrefhttps://github.com/econ-ark/econ-ark.org team members who helped with the project, particularly Alan Lujan, Akshay Shanker, and Matthew White (alpabetically).

0

1 Introduction

This trimmed version preserves the full section structure while removing most narrative detail.

1.1 Related literature

The full literature review is omitted in this lightweight scaffold.

2 Model

We retain one representative equation for compilation and structure checks:

$$C_t = \beta \, \mathbb{E}_t[C_{t+1}] + \varepsilon_t. \tag{1}$$

3 Parameterizing the model

Calibration details are intentionally abbreviated.

4 Comparing fiscal stimulus policies

Most policy comparison text is removed; a minimal table is kept.

Table 1 Illustrative policy comparison

Policy	Spending effect	Welfare effect
Stimulus check	Medium	Medium
UI extension	High	High
Tax cut	Low	Low

5 Robustness in a HANK and SAM Model

Detailed robustness analysis is omitted.

Placeholder figure retained to satisfy minimal artifact requirements.

Figure 1 Placeholder robustness figure

6 Conclusion

Main takeaways are intentionally shortened for the stripped document version.

Appendices

A Appendix: Minimal retained content

This stripped version keeps one appendix section while removing most appendix material.

References

Auclert, Adrien, Matthew Rognlie, and Ludwig Straub (2020): "Micro Jumps, Macro Humps: Monetary Policy and Business Cycles in an Estimated HANK Model," Discussion paper, Stanford University, Revise and resubmit at American Economic Review.

Bardoczy, Bence (2024): "Spousal Insurance and the Amplification of Business Cycles," Manuscript, Board of Governors of the Federal Reserve System.

Blanchard, Olivier, and Jordi Galí (2010): "Labor Markets and Monetary Policy: A New Keynesian Model with Unemployment," *American Economic Journal: Macroeconomics*, 2(2), 1–30.

Carroll, Christopher D, Edmund Crawley, Jiri Slacalek, and Matthew N White (2020): "Modeling the consumption response to the CARES act," *Covid Economics*, 10, 62–86.

Christiano, Lawrence J., Martin S. Eichenbaum, and Mathias Trabandt (2016): "Unemployment and Business Cycles," *Econometrica*, 84(4), 1523–1569.

Den Haan, Wouter J, Kenneth L Judd, and Michel Juillard (2010): "Computational suite of models with heterogeneous agents: Incomplete markets and aggregate uncertainty," *Journal of Economic Dynamics and Control*, 34(1), 1–3.

den Haan, Wouter J., Garey Ramey, and Joel Watson (2000): "Job Destruction and Propagation of Shocks," *The American Economic Review*, 90(3), 482–498.

Fagereng, Andreas, Martin B. Holm, and Gisle J. Natvik (2021): "MPC Heterogeneity and Household Balance Sheets," *American Economic Journal: Macroeconomics*, 13(4), 1–54.

Gornemann, Nils, Keith Kuester, and Makoto Nakajima (2021): "Doves for the Rich, Hawks for the Poor? Distributional Consequences of Systematic Monetary Policy," ECONtribute Discussion Papers Series 089, University of Bonn and University of Cologne, Germany.

Ravn, Morten O., and Vincent Sterk (2017): "Job uncertainty and deep recessions," *Journal of Monetary Economics*, 90, 125–141.

Ravn, Morten O, and Vincent Sterk (2021): "Macroeconomic Fluctuations with HANK & SAM: an Analytical Approach," *Journal of the European Economic Association*, 19, 1162–1202.

Rotemberg, Julio J. (1982): "Sticky Prices in the United States," *Journal of Political Economy*, 90(6), 1187–1211.

Rothstein, Jesse, and Robert G Valletta (2017): "Scraping by: Income and program participation after the loss of extended unemployment benefits," *Journal of Policy Analysis and Management*, 36(4), 880–908.

Silva, José, and Manuel Toledo (2009): "Labor Turnover Costs and The Cyclical Behavior or Vacancies and Unemployment," *Macroeconomic Dynamics*, 13(S1), 76–96.